

Program : Diploma in Civil and Environmental Engineering	
Course Code: 6359	Course Title: Environmental Engineering Drawing Lab.
Semester: 6	Credits: 1.5
Course Category: Programme Elective Course	
Periods per week: 3 (L:1 T:0 P:2)	Periods per semester: 45

Course Objectives:

- To enable students to identify and sketch water treatment units.
- To enable students to identify and sketch sewage treatment units.

Course Prerequisites:

Topic	Course code	Course name	Semester
Water treatment		Environmental Engineering	5
Sewage treatment and disposal		Environmental Engineering	5
Visit to water treatment plant and sewage treatment plant		Environmental Engineering lab	5

Course Outcomes:

On completion of the course, the student will be able to:

CO _n	Description	Duration (Hours)	Cognitive level
CO1	Prepare drawing of treatment units to effect sedimentation of raw water.	9	Applying
CO2	Prepare drawing of filtration units to treat raw water.	6	Applying
CO3	Prepare drawing of treatment units of a municipal sewage treatment plant.	14	Applying
CO4	Prepare drawings of septic tank and sludge digester.	9	Applying
	Open ended experiments	3	Applying
	Lab tests	4	Applying

CO-PO Mapping:

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	3		2				
CO2	3		2				
CO3	3		2				
CO4	3		2				

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline

Module Outcomes	Description	Duration (Hours)	Cognitive Level
CO1	Prepare drawing of treatment units to effect sedimentation of raw water.		
M1.01	Prepare layout of a water treatment plant to treat surface water and to treat ground water.	3	Understanding
M1.02	Prepare plan and sectional views of a sedimentation tank (any one type) for the given data.	3	Applying
M1.03	Prepare drawings (plan and sectional view) of flash mixer and clariflocculator.	3	Applying
CO2	Prepare drawing of filtration units to treat raw water.		
M2.01	Sketch plan and sectional views of a rapid sand filter and slow sand filter for the given data.	6	Applying
	Lab test 1	2	
CO3	Prepare drawing of treatment units of a municipal sewage treatment plant.		
M3.01	Prepare layout of a typical municipal sewage treatment plant.	2	Understanding
M3.02	Prepare plan and sectional view of a screen chamber for the given data.	3	Applying
M3.03	Prepare drawing of grit chamber for the given data.	3	Applying

M3.04	Prepare drawing (plan and section) of primary sedimentation tank for the given data.	3	Applying
M3.05	Prepare process flow diagrams of secondary treatment of sewage through trickling filters and activated sludge process.	3	Understanding
CO4	Prepare drawings of septic tank and sludge digester.		
M4.01	Prepare plan and sectional view of a septic tank and effluent disposal arrangement for the given data.	3	Applying
M4.02	Prepare sketch of sludge digester and sludge drying beds.	3	Applying
M4.03	Prepare sketch of a biogas plant.	3	Applying
M4.04	Open ended experiments	3	Applying
	Lab test- II	2	

Text / Reference:

T/R	Book Title/Author
T1	S.C.Rangwala, Water Supply and Sanitary Engineering, Charotar Publishing House
T2	Garg, S.K., Environmental Engineering Vol. I and Vol. II, Khanna Publishers
T3	K.N.Dugal, Elements of Public Health Engineering, S.Chand&Co.
T4	Sharma S.C, Environmental Engineering, Khanna Publishing House, New Delhi
R1	Metcalf & Eddy, Wastewater engineering: treatment and reuse, Tata McGraw Hill
R2	Birdie, G. S. and Birdie, J.S. Water Supply and Sanitary Engineering, Dhanpat Rai Publications

Online resources:

Sl.No.	Website Link
1	http://nptel.ac.in/courses